

CHAPTER 9

Biotechnology: Principles and Processes

Principles of Biotechnology

- In a chromosome, there is a specific DNA sequence, responsible for initiating replication. It is called as: (2024 Re)
 - recognition sequence
 - cloning site
 - restriction site
 - ori* site
- What is the fate of a piece of DNA carrying a gene of interest which is transferred into an alien organism? (2024)
 - The piece of DNA would be able to multiply, itself independently in the progeny cells of the organism.
 - It may get integrated into the genome of recipient.
 - It may multiply and be inherited along with the host DNA.
 - The alien piece of DNA is not an integral part of chromosome.
 - It shows ability to replicate.

Choose the correct answer from the options given below:

 - B and C only
 - A and E only
 - A and B only
 - D and E only
- Pathogenic bacteria gain resistance to antibiotics due to changes in their: (2022 Re)
 - Nucleoid
 - Cosmids
 - Plasmids
 - Nucleus
- Which of the following methods is not commonly used for introducing foreign DNA into the plant cell? (2022 Re)
 - Bacteriophages
 - Agrobacterium* mediated transformation
 - Gene gun
 - 'Disarmed pathogen' vectors
- The cutting of DNA at specific locations became possible with the discovery of: (2015 Re)
 - Probes
 - Selectable markers
 - Ligases
 - Restriction enzymes

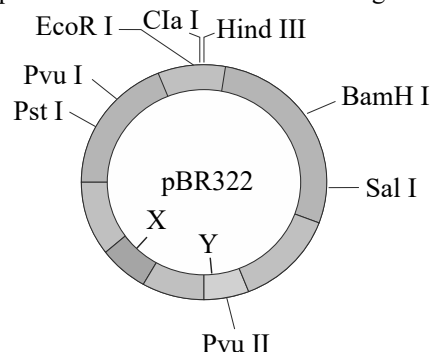
Tools of Recombinant DNA Technology

- Identify the **incorrect** statement related to gel electrophoresis. (2024 Re)
 - Separated DNA fragments can be directly seen under UV radiation.
 - Separated DNA can be extracted from gel piece.
 - Fragment of DNA moves toward anode.
 - Sieving effect of agarose gel helps in separation of DNA fragments.

- Recombinant DNA molecule can be created normally by cutting the vector DNA and source DNA respectively with: (2024 Re)
 - Hind* II, *Hind* II
 - Hind* II, *Alu* I
 - Hind* II, *Eco*R I
 - Hind* II, *Bam*H I
- Select the restriction endonuclease enzymes whose restriction sites are present for the tetracycline resistance (*tet*^R) gene in the pBR322 cloning vector. (2024 Re)
 - Bam* HI and *Sal* I
 - Sal* I and *Pst* I
 - Pst* I and *Pvu* I
 - Pvu* I and *Bam* HI
- Which of the following are correct about *Eco*RI? (2024 Re)
 - Cut the DNA with blunt end
 - Cut the DNA with sticky end
 - Recognises a specific palindromic sequence
 - Cut the DNA between the base G and A when encounters the DNA sequence 'GAATTC'
 - Exonuclease

Choose the **correct** answer from the options given below:

 - B, C, E only
 - A, D, E only
 - A, C, D only
 - B, C, D only
- Hind* II always cuts DNA molecules at a particular point called recognition sequence and it consists of: (2024)
 - 4 bp
 - 10 bp
 - 8 bp
 - 6 bp
- The following diagram showing restriction sites in *E. coli* cloning vector pBR322. Find the role of 'X' and 'Y' genes: (2024)



- The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.
- Gene 'X' is responsible for recognition sites and 'Y' is responsible for antibiotic resistance.
- The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.
- The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.

12. The “Ti plasmid” of *Agrobacterium tumefaciens* stands for
 a. Tumor inducing plasmid (2024)
 b. Temperature independent plasmid
 c. Tumour, inhibiting plasmid
 d. Tumor independent plasmid
13. Upon exposure to UV radiation, DNA stained with ethidium bromide will show (2023)
 a. Bright orange colour b. Bright red colour
 c. Bright blue colour d. Bright yellow colour
14. In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used. (2023)
 a. Silver b. Copper
 c. Zinc d. Tungsten or gold
15. Which of the following is not a cloning vector? (2023)
 a. Probe b. BAC c. YAC d. pBR322
16. Ligation of foreign DNA at which of the following site will result in loss of tetracycline resistance of pBR322: (2023-Manipur)
 a. *Pst* I b. *Pvu* I c. *Eco*R I d. *Bam*H I
17. Match List-I with List-II. (2023-Manipur)

List-I		List-II	
(A)	Kanamycin	(I)	Delivers genes into animal cells
(B)	<i>Clal</i>	(II)	Selectable marker
(C)	Disarmed retroviruses	(III)	Restriction site
(D)	Kanamycin ^R gene	(IV)	Antibiotic resistance

Choose the **correct** answer from the options given below:

- a. A-II, B-III, C-I, D-IV b. A-III, B-I, C-IV, D-II
 c. A-IV, B-III, C-I, D-II d. A-II, B-IV, C-I, D-III
18. Which of the following can act as molecular scissors? (2023-Manipur)
 a. Restriction enzymes b. DNA ligase
 c. RNA polymerase d. DNA polymerase
19. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). (2022 Re)
Assertion (A): When a particular restriction enzyme cuts strand of DNA, overhanging stretches or sticky ends are formed.
Reason (R): Some restriction enzymes cut the strand of DNA a little away from the centre of the palindromic site.
 In the light of the above statements, choose the correct answer from the options given below
 a. (A) is not correct but (R) is correct
 b. Both (A) and (R) are correct and (R) is the correct explanation of (A)
 c. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 d. (A) is correct but (R) is not correct
20. Separation of DNA fragments is done by a technique known as (2022 Re)
 a. Gel electrophoresis b. Polymerase Chain Reaction
 c. Recombinant technology d. Southern blotting

21. In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme? (2022)

- a. 5' G T A T T C 3' ; 3' C A T A A G 5'
 b. 5' G A T A C T 3' ; 3' C T A T G A 5'
 c. 5' G A A T T C 3' ; 3' C T T A A G 5'
 d. 5' C T C A G T 3' ; 3' G A G T C A 5'

22. Which of the following is not a desirable feature of a cloning vector? (2022)

- a. Presence of two or more recognition sites
 b. Presence of origin of replication
 c. Presence of a marker gene
 d. Presence of single restriction enzyme site

23. Given below are two statements:

Statement I: Restriction endonucleases recognise specific sequence to cut DNA known as palindromic nucleotide sequence.

Statement II: Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site

In the light of the above statements, choose the most appropriate answer from the options given below.

- a. Statement I is incorrect but Statement II is correct
 b. Both Statement I and Statement II are correct
 c. Both statement I and statement II are incorrect
 d. Statement I is correct but Statement II is incorrect

24. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as: (2021)

- a. Bright orange bands b. Dark red bands
 c. Bright blue bands d. Yellow bands

25. Plasmid pBR322 has *Pst*I restriction enzyme site within gene *amp^R* that confers ampicillin resistance. If this enzyme is used for inserting a gene for β -galactoside production and the recombinant plasmid is inserted in an *E.coli* strain (2021)

- a. The transformed cells will have the ability to resist ampicillin as well as produce β -galactoside.
 b. It will lead to lysis of host cell.
 c. It will be able to produce a novel protein with dual ability.
 d. It will not be able to confer ampicillin resistance to host cell.

26. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is: (2021)

- a. Okazaki sequences
 b. Palindromic Nucleotide sequences
 c. Poly (A) tail sequences
 d. Degenerate primer sequence

27. First discovered restriction endonuclease that always cuts DNA molecule at a particular point by recognising a specific sequence of six base pairs is: (2020-Covid)

- a. Adenosine deaminase
 b. Thermostable DNA polymerase
 c. *Hind* II
 d. *Eco*RI

28. Match the organism with its use in biotechnology. (2020)

1.	<i>Bacillus thuringiensis</i>	(i)	Cloning vector
2.	<i>Thermus aquaticus</i>	(ii)	Construction of first rDNA molecule
3.	<i>Agrobacterium tumefaciens</i>	(iii)	DNA polymerase
4.	<i>Salmonella typhimurium</i>	(iv)	Cry proteins

Select the correct option from the following

- a. 1-(iv); 2-(iii); 3-(i); 4-(ii)
 b. 1-(iii); 2-(ii); 3-(iv); 4-(i)
 c. 1-(iii); 2-(iv); 3-(i); 4-(ii)
 d. 1-(ii); 2-(iv); 3-(iii); 4-(i)

29. The sequence that controls the copy number of the linked DNA in the vector, is termed: (2020)

- a. *Ori* site b. Palindromic sequence
 c. Recognition site d. Selectable marker

30. The specific palindromic sequence which is recognized by *EcoRI* is: (2020)

- a. 5' - GGAACC - 3' b. 5' - CTTAAG - 3'
 3' - CCTTGG - 5' 3' - GAATTC - 5'
 c. 5' - GGATCC - 3' d. 5' - GAATTC - 3'
 3' - CCTAGG - 5' 3' - CTTAAG - 5'

31. Identify the wrong statement with regard to restriction enzymes. (2020)

- a. They cut the strand of DNA at palindromic sites.
 b. They are useful in genetic engineering.
 c. Sticky ends can be joined by using DNA ligases.
 d. Each restriction enzyme functions by inspecting the length of a DNA sequence.

32. Choose the correct pair from the following: (2020)

a.	Polymerases	-	Break the DNA into fragments
b.	Nucleases	-	Separate the two strands of DNA
c.	Exonucleases	-	Make cuts at specific positions within DNA
d.	Ligases	-	Join the two DNA molecules

33. Following statements describe the characteristics of the enzyme restriction endonuclease. Identify the incorrect statement. (2019)

- a. The enzyme cuts DNA molecule at identified position within the DNA.
 b. The enzyme binds DNA at specific sites and cuts only one of the two strands.
 c. The enzyme cuts the sugar-phosphate backbone at specific sites on each strand.
 d. The enzyme recognizes a specific palindromic nucleotide sequence in the DNA.

34. A gene whose expression helps to identify transformed cell is known as (2017-Delhi)

- a. Selectable marker b. Vector
 c. Plasmid d. Structural gene

35. Restriction endonucleases are: (2017-Gujarat)

- a. Used in genetic engineering for ligating two DNA molecules
 b. Used for in vitro DNA synthesis
 c. Synthesised by bacteria as part of their defense mechanism
 d. Present in mammalian cell for degradation of DNA when the cell dies

36. Which of the following restriction enzymes produces blunt ends? (2016 - II)

- a. *Xho* I b. *Hind* III c. *Sal* I d. *Eco* RV

37. A foreign DNA and plasmid cut by the same restriction endonuclease can be joined to form a recombinant plasmid using: (2016 - II)

- a. Polymerase-III b. Ligase
 c. *Eco* RI d. Taq polymerase

38. Which of the following is a restriction endonuclease?

(2016 - I)

- a. *Hind* II b. Protease c. DNase I d. RNase

39. Which of the following is not a feature of the plasmids? (2016 - I)

- a. Independent replication b. Circular structure
 c. Transferable d. Single-stranded

40. The DNA molecule to which the gene of interest is integrated for cloning is called: (2015 Re)

- a. Vector b. Template
 c. Carrier d. Transformer

41. The introduction of t-DNA into plants involves: (2015 Re)

- a. Altering the pH of the soil, then heat shocking the plants
 b. Exposing the plants to cold for a brief period
 c. Allowing the plant roots to stand in water
 d. Infection of the plant by *Agrobacterium tumefaciens*

42. Which vector can clone only a small fragment of DNA? (2014)

- a. Cosmid
 b. Bacterial artificial chromosome
 c. Yeast artificial chromosome
 d. Plasmid

43. DNA fragments generated by the restriction endonuclease in a chemical reaction can be separated by: (2013)

- a. Restriction mapping
 b. Centrifugation
 c. Polymerase chain reaction
 d. Electrophoresis

44. The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of: (2013)

- a. Inactivation of glycosidase enzyme in recombinant bacteria
 b. Non-recombinant bacteria containing beta-galactosidase
 c. Insertional inactivation of alpha-galactosidase in non-recombinant bacteria
 d. Insertional inactivation of beta-galactosidase in recombinant bacteria

Processes of Recombinant DNA Technology

45. Following are the steps involved in the process of PCR. (2024 Re)

- A. Annealing
- B. Amplification (~1 billion times)
- C. Denaturation
- D. Treatment with *Taq* polymerase and deoxynucleotides
- E. Extension

Choose the correct sequence of steps of PCR from the options given below:

- a. C→A→D→E→B b. A→B→E→D→C
- c. A→C→E→D→B d. D→B→E→C→A

46. Which of the following statements is incorrect? (2024)

- a. Bio-reactors are used to produce small scale bacterial cultures.
- b. Bio-reactors have an agitator system, an oxygen delivery system and foam control system.
- c. A bio-reactor provides optimal growth conditions for achieving the desired product.
- d. Most commonly used bio-reactors are of stirring type.

47. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out (2023)

- a. Polysaccharides b. RNA
- c. DNA d. Histones

48. Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence (2023)

- A. Insertion of recombinant DNA into the host cell.
- B. Cutting of DNA at specific location by restriction enzyme.
- C. Isolation of desired DNA fragment.
- D. Amplification of gene of interest using PCR.

Choose the correct answer from the option given below:

- a. B, D, A, C b. B, C, D, A
- c. C, A, B, D d. C, B, D, A

49. Thermostable DNA polymerase used in PCR was isolated from: (2023-Manipur)

- a. *Thermus aquaticus*
- b. *Escherichia coli*
- c. *Agrobacterium tumefaciens*
- d. *Bacillus thuringiensis*

50. The enzyme (a) is needed for isolating genetic material from plant cells and enzyme (b) for isolating genetic material from fungus. Choose the correct pair of options from the following: (2022 Re)

- a. (a) Cellulase (b) Lipase
- b. (a) Cellulase (b) Protease
- c. (a) Cellulase (b) Chitinase
- d. (a) Chitinase (b) Lipase

51. Refer to the following statements for agarose gel electrophoresis: (2022 Re)

- A. Agarose is a natural polymer obtained from sea-weed.
- B. The separation of DNA molecules in agarose-gel electrophoresis depends on the size of DNA.

C. The DNA migrates from negatively-charged electrode to the positively-charged electrode

D. The DNA migrates from positively-charged electrode to the negatively-charged electrode.

Choose the **most appropriate** answer from the options given below

- a. (B), (C) and (D) only b. (A) and (B) only
- c. (A), (B) and (C) only d. (A), (B) and (D) only

52. Which one of the following statement is **not true** regarding gel electrophoresis technique? (2022)

- a. Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.
- b. The process of extraction of separated DNA strands from gel is called elution.
- c. The separated DNA fragments are stained by using ethidium bromide.
- d. The presence of chromogenic substrate gives blue coloured DNA bands on the gel

53. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). (2022)

Assertion (A): Polymerase chain reaction is used in DNA amplification

Reason (R): The ampicillin resistant gene is used as a selectable marker to check transformation

In the light of the above statements, choose the correct answer from the options given below.

- a. (A) is not correct but (R) is correct
- b. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- c. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- d. (A) is correct but (R) is not correct

54. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out: (2021)

- a. DNA b. Histones
- c. Polysaccharides d. RNA

55. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first? (2021)

- a. Extension b. Denaturation
- c. Ligation d. Annealing

56. In gel electrophoresis, separated DNA fragments can be visualized with the help of: (2020)

- a. Ethidium bromide in UV radiation
- b. Acetocarmine in UV radiation
- c. Ethidium bromide in infrared radiation
- d. Acetocarmine in bright blue light

57. In a mixture, DNA fragments are separated by (2020-Covid)

- a. Restriction digestion
- b. Electrophoresis
- c. Polymerase chain reaction
- d. Bioprocess engineering

58. In recombinant DNA technology antibiotics are used: (2020-Covid)

- To detect alien DNA
- To impart disease-resistance to the host plant
- As selectable markers
- To keep medium bacteria-free

59. Match the following techniques or instruments with their usage: (2020-Covid)

1. Bioreactor	(i)	Separation of DNA fragments
2. Electrophoresis	(ii)	Production of large quantities of products
3. PCR	(iii)	Detection of pathogen, based on antigen-antibody reaction
4. ELISA	(iv)	Amplification of nucleic acids

Select the correct option from following:

- | | | | | |
|----|-------|-------|-------|-------|
| | (1) | (2) | (3) | (4) |
| a. | (ii) | (i) | (iv) | (iii) |
| b. | (iv) | (iii) | (ii) | (i) |
| c. | (ii) | (i) | (iii) | (iv) |
| d. | (iii) | (ii) | (iv) | (i) |

60. Spooling is: (2020-Covid)

- Cutting of separated DNA bands from the agarose gel
- Transfer of separated DNA fragments to synthetic membranes
- Collection of isolated DNA
- Amplification of DNA

61. Select the correct statement from the following: (2020-Covid)

- The polymerase enzyme joins the gene of interest and the vector DNA
- Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzymes of optimum conditions
- PCR is used for isolation and separation of gene of interest
- Gel electrophoresis is used for amplification of a DNA segment

62. Which one of the following equipments is essentially required for growing microbes on a large scale, for industrial production of enzymes? (2019)

- BOD incubator
- Sludge digester
- Industrial oven
- Bioreactor

63. DNA precipitation out of a mixture of biomolecules can be achieved by treatment with (2019)

- Isopropanol
- Chilled ethanol

- Methanol at room temperature
- Chilled chloroform

64. The correct order of steps in Polymerase Chain Reaction (PCR) is: (2018)

- Extension, Denaturation, Annealing
- Annealing, Extension, Denaturation
- Denaturation, Extension, Annealing
- Denaturation, Annealing, Extension

65. What is the criterion for DNA fragments movement on agarose gel during gel electrophoresis? (2017-Delhi)

- The larger the fragment size, the farther it moves
- The smaller the fragment size, the farther it moves
- Positively charged fragments move to farther end
- Negatively charged fragments do not move

66. The process of separation and purification of expressed protein before marketing is called (2017-Delhi)

- Upstream processing
- Downstream processing
- Bioprocessing
- Postproduction processing

67. The DNA fragments separated on an agarose gel can be visualised after staining with: (2017-Delhi)

- Bromophenol blue
- Acetocarmine
- Aniline blue
- Ethidium bromide

68. Which of the following is not a component of downstream processing? (2016 - II)

- Preservation
- Expression
- Separation
- Purification

69. Stirred-tank bioreactors have been designed for: (2016 - II)

- Availability of oxygen throughout the process
- Ensuring anaerobic conditions in the culture vessel
- Purification of product
- Addition of preservatives to the product

70. The *Taq* polymerase enzyme is obtained from: (2016 - I)

- Thermus aquaticus*
- Thiobacillus ferrooxidans*
- Bacillus subtilis*
- Pseudomonas putida*

71. *In vitro* clonal propagation in plants is characterised by: (2014)

- Microscopy
- PCR and RAPD
- Northern blotting
- Electrophoresis and HPLC

72. Which of the following is not correctly matched for the organism and its cell wall degrading enzyme? (2013)

- Fungi - Chitinase
- Bacteria - Lysozyme
- Plant cells - Cellulase
- Algae - Methyrase

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (a) | 3. (c) | 4. (a) | 5. (d) | 6. (a) | 7. (a) | 8. (a) | 9. (d) | 10. (d) |
| 11. (d) | 12. (a) | 13. (a) | 14. (d) | 15. (a) | 16. (d) | 17. (c) | 18. (a) | 19. (b) | 20. (a) |
| 21. (c) | 22. (a) | 23. (b) | 24. (a) | 25. (d) | 26. (b) | 27. (c) | 28. (a) | 29. (a) | 30. (d) |
| 31. (c) | 32. (d) | 33. (b) | 34. (a) | 35. (c) | 36. (d) | 37. (b) | 38. (a) | 39. (d) | 40. (a) |
| 41. (d) | 42. (d) | 43. (d) | 44. (b) | 45. (a) | 46. (a) | 47. (c) | 48. (b) | 49. (a) | 50. (c) |
| 51. (c) | 52. (d) | 53. (c) | 54. (a) | 55. (b) | 56. (a) | 57. (b) | 58. (c) | 59. (a) | 60. (c) |
| 61. (b) | 62. (d) | 63. (b) | 64. (d) | 65. (b) | 66. (b) | 67. (d) | 68. (b) | 69. (a) | 70. (a) |
| 71. (b) | 72. (d) | | | | | | | | |

